Project: Worked examples
Title: Bolted joint (simplified VDI 2230)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

## 1. Scope and method

One bolt clamping members under an axial working load: the preload from the tightening torque with its scatter (or as measured), the stiffness of bolt and members, how much of the working load the bolt sees, the residual clamp after embedding, separation, yield under preload plus working load, the thread engagement needed against stripping, and the fatigue amplitude.
$A_s = (\pi/4)\cdot((d_2 + d_3)/2)^2$, $d_2 = d - 0.6495\,P$, $d_3 = d - 1.2269\,P$. Nut factor: $F_M = M_A/(K\,d)$. Friction: $F_M = M_A/(0.16\,P + 0.58\,d_2\,\mu_G + 0.5\,D_{Km}\,\mu_K)$. $F_{M,min} = F_M/\alpha_A$. $\delta_S = (0.4\,d + 0.4\,d)/(E_S\,A_N) + l_k/(E_S\,A_s) + 0.5\,d/(E_S\,A_{d3})$. Members: two frusta, $k = 0.5774\,\pi\,E\,d\,/\,\ln[((1.155\,t + d_w - d_h)(d_w + d_h))/((1.155\,t + d_w + d_h)(d_w - d_h))]$. $\Phi = n\,k_S/(k_S + k_P)$. $F_Z = f_Z\,(k_S\,k_P)/(k_S + k_P)$. Given preload: $F_M = F_V$; $k_S$, $k_P$ and $A_s$ may be given (measured, FE, a fastener record) instead of derived. $F_{KR} = \max(F_{M,min} - (1 - \Phi)\,F_A - F_Z, 0)$. $F_{S,max} = \max(F_M + \Phi\,F_A, F_A)$: once the joint opens the bolt carries the whole working load. $\sigma_{red} = \sqrt{(\sigma_z^2 + 3\,(0.5\,\tau)^2)}$. $\sigma_a = (F_{S,max} - F_M)/(2\,A_s)$. $m_{min} = S\,F_{S,max}/(0.5\,\pi\,d\,\tau_B)$.
Ref: VDI 2230 Part 1:2015 §5.1–§5.5 — Systematic calculation of highly stressed bolted joints — calculation steps R0–R13
Ref: ISO 898-1:2013 §9.1.6 — Tensile stress area of the thread
Ref: Shigley's Mechanical Engineering Design §8-5, §8-7 — Joint stiffness (frusta); preload and torque

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| d | Nominal thread diameter | 12 | mm | drawing: DRG-FLG-02 rev A | 3 … 64 mm |
| P | Thread pitch | 1.75 | mm | spec: ISO 261 coarse pitch | 0.35 … 6 mm |
| R_p02 | Bolt 0.2 % proof (yield) strength | 640 | MPa | spec: ISO 898-1 class 8.8 | 200 … 1200 MPa |
| E_S | Bolt Young's modulus | 205 | GPa | spec: steel | 60 … 250 GPa |
| sigma_A | Bolt endurance limit (stress amplitude) | 50 | MPa | assumption: Rolled-before-heat-treatment M12, VDI 2230 order of magnitude. | 10 … 200 MPa |
| l_k | Clamp length | 40 | mm | drawing: DRG-FLG-02 rev A | 1 … 2000 mm |
| d_h | Hole diameter | 13 | mm | drawing: DRG-FLG-02 rev A | 1 … 100 mm |
| d_w | Head / washer bearing diameter | 18 | mm | spec: ISO 4762 head bearing | 2 … 200 mm |
| E_P | Clamped members' Young's modulus | 205 | GPa | spec: steel | 5 … 250 GPa |
| n | Load introduction factor (0 = at the clamped faces, 1 = under the head) | 0.5 | | assumption: Load introduced at mid-thickness. | 0 … 1 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| f_Z | Embedding (settling) loss | 11 | µm | assumption: Three interfaces, machined. | 0 … 100 µm |
| method | Preload method | friction | | spec: Assembly procedure AP-7 | any |
| M_A | Tightening torque | 80 | N*m | spec: Assembly procedure AP-7 | 0.1 … 10000 N*m |
| K | Nut factor (nut-factor method) | 0.2 | | assumption: Not used by the friction method. | 0.1 … 0.35 |
| mu_G | Thread friction coefficient | 0.12 | | assumption: Lightly oiled zinc-plated thread. | 0.06 … 0.25 |
| mu_K | Head / nut bearing friction coefficient | 0.12 | | assumption: Lightly oiled head bearing. | 0.06 … 0.25 |
| alpha_A | Tightening scatter factor (F_M,max / F_M,min) | 1.6 | | assumption: Torque wrench, VDI 2230 Table A8 order. | 1 … 4 |
| F_A | Axial working load (per bolt) | 60 | kN | spec: LOAD-CASE-3 | 0 … 5000 kN |
| F_KR_req | Required residual clamp force | 5 | kN | spec: SPEC-FLG §6 sealing | 0 … 5000 kN |
| tau_B | Nut / internal-thread material shear strength | 400 | MPa | assumption: Nut material shear strength. | 50 … 1200 MPa |
| m_eff | Thread engagement length provided | 12 | mm | drawing: DRG-FLG-02 rev A | 1 … 200 mm |
| S_strip | Safety factor on stripping | 1.25 | | assumption: Stripping safety factor. | 1 … 3 |

## 3. Assumptions

A1. The joint is concentrically clamped and concentrically loaded; the members behave as two 30° frusta in series. (Eccentric clamping and loading (VDI 2230 full procedure) are outside this sheet.)
A2. Thread torsion is taken at 50 % in the equivalent stress for the working state. (Common VDI 2230 practice (k_τ = 0.5).)
A3. The working load is a pulsating 0…F_A load; the fatigue amplitude is Φ F_A / (2 A_s). (Compare with the bolt's endurance limit as entered.)
A4. Thread stripping uses a shear area of half the pitch per unit length on the nominal diameter. (A conservative simplification of ISO 898-2 / VDI 2230 §5.5.5.)
A5. sigma_A (Bolt endurance limit (stress amplitude)) is taken as 50 MPa. (Rolled-before-heat-treatment M12, VDI 2230 order of magnitude.)
A6. n (Load introduction factor (0 = at the clamped faces, 1 = under the head)) is taken as 0.5. (Load introduced at mid-thickness.)
A7. f_Z (Embedding (settling) loss) is taken as 11 µm. (Three interfaces, machined.)
A8. K (Nut factor (nut-factor method)) is taken as 0.2. (Not used by the friction method.)
A9. mu_G (Thread friction coefficient) is taken as 0.12. (Lightly oiled zinc-plated thread.)
A10. mu_K (Head / nut bearing friction coefficient) is taken as 0.12. (Lightly oiled head bearing.)

A11. alpha_A (Tightening scatter factor (F_M,max / F_M,min)) is taken as 1.6. (Torque wrench, VDI 2230 Table A8 order.)
A12. tau_B (Nut / internal-thread material shear strength) is taken as 400 MPa. (Nut material shear strength.)
A13. S_strip (Safety factor on stripping) is taken as 1.25. (Stripping safety factor.)

## 4. Working

[23] d_2 = d - 0.6495 * P — ISO 898-1 §9.1.6
= 12 mm − 0.6495 × 1.75 mm
= 10.863 mm
Pitch diameter

[24] d_3 = d - 1.2269 * P — ISO 898-1 §9.1.6
= 12 mm − 1.2269 × 1.75 mm
= 9.8529 mm
Minor diameter

[25] A_s = pi / 4 * ((d_2 + d_3) / 2)^2 — ISO 898-1 §9.1.6
= π / 4 × ((10.863 mm + 9.8529 mm) / 2)^2
= 84.266 mm²
Tensile stress area

[26] A_N = pi / 4 * d^2
= π / 4 × (12 mm)^2
= 113.1 mm²
Nominal shank area

[27] A_d3 = pi / 4 * d_3^2
= π / 4 × (9.8529 mm)^2
= 76.247 mm²
Minor-diameter area

[28] D_Km = (d_w + d_h) / 2
= (18 mm + 13 mm) / 2
= 15.5 mm
Mean bearing diameter under the head

[29] F_M = M_A / (0.16 * P + 0.58 * d_2 * mu_G + 0.5 * D_Km * mu_K) — VDI 2230 §5.4.3
= 80 N·m / (0.16 × 1.75 mm + 0.58 × 10.863 mm × 0.12 + 0.5 × 15.5 mm × 0.12)
= 40.69 kN
Assembly preload (friction)

[30] F_M_min = F_M / alpha_A — VDI 2230 R2
= 40.69 kN / 1.6
= 25.431 kN
Minimum preload after scatter

[31] sigma_M = F_M / A_s
= 40.69 kN / 84.266 mm²
= 482.87 MPa
Preload stress

[32] M_G = F_M * (0.16 * P + 0.58 * d_2 * mu_G) — VDI 2230 §5.4.3
= 40.69 kN × (0.16 × 1.75 mm + 0.58 × 10.863 mm × 0.12)
= 42.158 N·m
Thread torque under preload

[33] W_p = pi / 16 * d_3^3
= π / 16 × (9.8529 mm)^3
= 187.81 mm³
Polar section modulus at the minor diameter

[34] tau_M = M_G / W_p
= 42.158 N·m / 187.81 mm³
= 224.47 MPa

Thread torsional stress under preload

[35] sigma_red_M = sqrt(sigma_M^2 + 3 * tau_M^2) VDI 2230 R8

= sqrt((482.87 MPa)^2 + 3 × (224.47 MPa)^2)

= 619.94 MPa

Equivalent stress at assembly (full torsion)

[36] nu_M = sigma_red_M / R_p02

= 619.94 MPa / 640 MPa

= 0.96866

Utilisation of proof strength at assembly

[37] delta_S = 0.4 * d / (E_S * A_N) + l_k / (E_S * A_s) + 0.5 * d / (E_S * A_d3) + 0.4 * d / VDI 2230 Sheet 1.1

= 0.4 × 12 mm / (205 GPa × 113.1 mm²) + 40 mm / (205 GPa × 84.266 mm²) + 0.5 × 12 mm / (205 G

76.247 mm²) + 0.4 × 12 mm / (205 GPa × 113.1 mm²)

= 0.0000031135 m/kN

Bolt resilience (head, shank, engaged thread, nut)

[38] k_S = 1 / delta_S

= 1 / 0.0000031135 m/kN

= 321190 kN/m

Bolt stiffness

[39] t_f = l_k / 2

= 40 mm / 2

= 20 mm

Frustum height (two equal frusta)

[40] k_f = 0.5774 * pi * E_P * d / ln(((1.155 * t_f + d_w - d_h) * (d_w + d_h)) / ((1.155 * t

(d_w - d_h))) Shigley eq. 8-20

= 0.5774 × π × 205 GPa × 12 mm / ln((1.155 × 20 mm + 18 mm − 13 mm) × (18 mm + 13 mm) / ((1.1

mm + 18 mm + 13 mm) × (18 mm − 13 mm)))

= 3815600 kN/m

Stiffness of one frustum

[41] k_P = k_f / 2 Shigley eq. 8-21

= 3815600 kN/m / 2

= 1907800 kN/m

Clamped members' stiffness (two frusta in series)

[42] Phi = n * k_S / (k_S + k_P) VDI 2230 R3

= 0.5 × 321190 kN/m / (321190 kN/m + 1907800 kN/m)

= 0.072047

Load factor

[43] F_Z = f_Z * (k_S * k_P) / (k_S + k_P) VDI 2230 R4

= 11 µm × 321190 kN/m × 1907800 kN/m / (321190 kN/m + 1907800 kN/m)

= 3.024 kN

Preload loss from embedding

[44] F_SA = Phi * F_A

= 0.072047 × 60 kN

= 4.3228 kN

Additional bolt load from the working load

[45] F_PA = (1 - Phi) * F_A

= (1 − 0.072047) × 60 kN

= 55.677 kN

Load relieved from the members

[46] F_KR = max(F_M_min - F_PA - F_Z, 0 kN) VDI 2230 R7

= max(25.431 kN − 55.677 kN − 3.024 kN, 0 kN)

= 0 kN

Residual clamp force (minimum; zero once the joint opens)

[47] F_S_max = max(F_M + F_SA, F_A)
= max(40.69 kN + 4.3228 kN, 60 kN)
= 60 kN
Maximum bolt load (maximum preload + additional load; the whole working load once the joint opens)
[48] sigma_z_max = F_S_max / A_s
= 60 kN / 84.266 mm²
= 712.03 MPa
Maximum bolt tensile stress
[49] sigma_red_B = sqrt(sigma_z_max^2 + 3 * (0.5 * tau_M)^2) VDI 2230 R8
= sqrt((712.03 MPa)^2 + 3 × (0.5 × 224.47 MPa)^2)
= 738.09 MPa
Equivalent stress in the working state (50 % torsion)
[50] u_yield = sigma_red_B / R_p02
= 738.09 MPa / 640 MPa
= 1.1533
Yield utilisation in the working state
[51] sigma_a = (F_S_max - F_M) / (2 * A_s) VDI 2230 R9
= (60 kN - 40.69 kN) / (2 × 84.266 mm²)
= 114.58 MPa
Fatigue stress amplitude (pulsating 0…F_A load)
[52] u_fatigue = sigma_a / sigma_A
= 114.58 MPa / 50 MPa
= 2.2916
Fatigue utilisation
[53] m_min = S_strip * F_S_max / (0.5 * pi * d * tau_B) VDI 2230 §5.5.5
= 1.25 × 60 kN / (0.5 × π × 12 mm × 400 MPa)
= 9.9472 mm
Thread engagement needed against stripping
[54] u_strip = m_min / m_eff
= 9.9472 mm / 12 mm
= 0.82893
Engagement utilisation
[55] F_A_sep = (F_M_min - F_Z) / (1 - Phi) VDI 2230 R7
= (25.431 kN - 3.024 kN) / (1 - 0.072047)
= 24.147 kN
Working load at which the members separate (minimum preload, after embedding)
[56] F_proof = R_p02 * A_s
= 640 MPa × 84.266 mm²
= 53.93 kN
Bolt load at the proof (yield) strength
[57] n_0 = F_A_sep / F_A
= 24.147 kN / 60 kN
= 0.40245
Load factor against separation (F_A,sep / F_A)
[58] F_A_yield = min((F_proof - F_M) / Phi, F_proof)
= min((53.93 kN - 40.69 kN) / 0.072047, 53.93 kN)
= 53.93 kN
Working load at which the bolt reaches its proof load (separation included)
[59] n_L = F_A_yield / F_A
= 53.93 kN / 60 kN
= 0.89884
Load factor against axial yield of the bolt, torsion excluded

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| n_0 | 0.40245 | Load factor against separation |
| n_L | 0.89884 | Load factor against axial yield |
| A_s | 84.266 mm² | Tensile stress area |
| F_M | 40.69 kN | Assembly preload (maximum) |
| F_M_min | 25.431 kN | Minimum preload |
| sigma_M | 482.87 MPa | Preload stress |
| nu_M | 0.96866 | Assembly utilisation of proof strength |
| k_S | 321190 kN/m | Bolt stiffness |
| k_P | 1907800 kN/m | Members' stiffness |
| Phi | 0.072047 | Load factor |
| F_Z | 3.024 kN | Embedding loss |
| F_SA | 4.3228 kN | Additional bolt load |
| F_KR | 0 kN | Residual clamp |
| F_A_sep | 24.147 kN | Separation load |
| F_S_max | 60 kN | Maximum bolt load |
| sigma_z_max | 712.03 MPa | Maximum bolt tensile stress |
| sigma_red_B | 738.09 MPa | Working equivalent stress |
| sigma_a | 114.58 MPa | Fatigue amplitude |
| m_min | 9.9472 mm | Engagement needed |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | No separation | 0 kN | > 0 kN | – | VDI 2230 R7 |
| FAIL | Residual clamp against requirement | 0 kN | ≥ 5 kN | – | VDI 2230 R7 |
| FAIL | Fatigue amplitude | u_fatigue = 2.2916 | u_fatigue ≤ 1 | 2.29 | VDI 2230 R9 |
| FAIL | Yield in the working state | u_yield = 1.1533 | u_yield ≤ 1 | 1.15 | VDI 2230 R8 |
| PASS | Thread stripping | u_strip = 0.82893 | u_strip ≤ 1 | 0.829 | VDI 2230 §5.5.5 |

## 7. Applicability

WARNING — preload-yield: M_A: the preload alone takes the bolt above 90 % of its proof strength.
WARNING — separation: F_KR: the residual clamp force is not positive; the joint separates under the working load and the bolt carries all of it.

## 8. Verification

Template mech.bolted-joint@1.1.0 Content hash efa946ca6a5170d9 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0